

CSE0612215: Database Management System

Part A: Course Information

1. **Program:** BSc. in CSE
2. **Course Code:** CSE0612215
3. **Course Title:** Database Management System
4. **Course Type (GED/Core Course/Electives):** Core Course
5. **Year/Level/Semester/Term:** 3rd year/ 1st Semester
6. **Academic Session:** Autumn 2026
7. **Section:** 3C
8. **Prerequisite Course (If any):** N/A
9. **Credit Value:** 3.00
10. **Contact Hours:** 3.00 hours/Week
11. **Class Schedule:** Sunday (Room: 506, Period: 08.00 AM – 10.00 AM)
12. **Counseling Schedule:** Wednesday (9.00 AM – 12.00 PM),
Thursday (8.00 AM – 11.00 AM)
13. **Course Teacher:** Md. Moradul Siddique, Designation: Lecturer, Room: 418(10), Email: moradul_siddique@uits.edu.bd, Cell: +8801621291844, Specialization: Machine Learning, Deep Learning, Computer Vision, Cybersecurity, Bioinformatics, Internet of Things (IoT) and Healthcare AI.
14. **Rationale of the Course:** This course introduces database design and creation using a DBMS product. Emphasis is on data dictionaries, normalization, data integrity, data modeling, and creation of simple tables, queries, reports, and forms. Upon completion, students should be able to design and implement normalized database structures by creating simple database tables, queries, reports, and forms.
15. **Course Objectives:**
 1. Describe the fundamental elements of relational database management systems.
 2. Explain the basic concepts of relational data model, entity-relationship model, relational database design, relational algebra and SQL.
 3. Design ER-models to represent simple database application scenarios
 4. Convert the ER-model to relational tables, populate relational database and formulate SQL queries on data.
 5. Improve the database design by normalization, Familiar with basic database storage structures and access techniques.

Part B: Skill Mapping

16. Course Outcome (COs)

No.	COs	Taxonomy	CP/ WP	CA/ EA	KP/ WK	AT	DM&A
CO1	Explain the fundamental concepts, architecture, and applications of relational database management systems in organizational and real-world environments.	Aff4	-	-	3	CT, MT, FE	Lecture and Discussion
CO2	Apply and execute SQL queries, analyze relational algebra expressions, and evaluate database storage, indexing, and	Aff4	-	-	3	MT, FE	Lecture and Discussion

	transaction management techniques for efficient database operations.						n
CO3	Design normalized relational databases by modeling real-world problems using ER diagrams, applying normalization principles, and integrating data standards, integrity constraints, and data dictionary concepts.	Aff6	1,3,6,7	-	3,5,6	FE, R, A	Lecture and Demonstration

*COs- Course Outcome; BT- Bloom's Taxonomy; CP/WP- Complex Problems; EA/WA- Complex Activities; AT- Assessment Tools; KP/WK- Knowledge Profile; DM&A: Delivery Methods & Activities; FE- Final Exam; CT- Class Test, MT- Mid Term, A- Assignment.

17. Mapping of COs with Program Outcomes (POs)

Course Outcomes (COs)	PO(a)	PO(b)	PO(c)	PO(d)	PO(e)	PO(f)	PO(g)	PO(h)	PO(i)	PO(j)	PO(k)	PO(l)
CO1	√											
CO2		√										
CO3			√									

Part C: Teaching Learning Approach

18. Course plan specifying contents, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs.

* Commencement of the semester (Autumn 2026): 22 July 2026 and Last class of the semester: 19 November 2026

Week	Topics	Suggested Activity and Teaching Strategy	Assessment Strategy	Corresponding COs
1	DBMS: Introduction, goals of DBMS, Application of database systems, purpose of database systems database vs file system. View of Data: data abstraction, instances and schemas, data models. Data Engineering Overview	Students will introduce themselves using socialized recitation; course outline and DBMS basics will be discussed interactively; examples on DBMS vs file system will be shared using multimedia.	Midterm	CO1
2	Database Languages: Procedural and declarative, DML and DDL, DCL, TCL, data storage and definition language, Consistency constraints specification, data dictionary.	Database languages will be explained through interactive lecture; examples of DML, DDL, DCL, TCL	Midterm	CO2

	Database Users and Administrators: database users and user interfaces, database administrator.	shown through multimedia; roles of DB users and DBA discussed with examples.		
3	Database Storage and Querying: storage manager, the query processor. Transaction Management: ACID properties, Database Architecture: two-tier and three-tier architecture. Relational Model: introduction, characteristics, structures of relational databases.	Storage manager, query processor, and ACID properties explained using multimedia; two-tier and three-tier architecture discussed with diagrams; relational model introduced through interactive discussion.	Midterm Class Test 1	CO1
4	Keys in DBMS: super key, candidate key, primary key, composite key and foreign key. Fundamental Relational-Algebra operations: select (unary), project (unary), rename (unary), Cartesian product (binary), union (binary), set-difference (binary).	Types of keys demonstrated with sample tables; basic relational algebra taught with examples on board/multimedia; students practice identifying keys through discussion.	Midterm	CO2
5	Additional Relational-Algebra Operations: set-intersection (binary), natural join (binary), division (binary), assignment (unary). Extended Relational Algebra Operations: generalized projection, aggregation functions, outer join. SQL: Parts of SQL, Domain Types in SQL, Basic Schema Definition in SQL, Basic Structure of SQL Queries.	Advanced relational algebra explained with visual examples; SQL basics and schema definition discussed using multimedia; sample problems practiced through guided discussion.	Midterm	CO2
6	SQL-Set Operations: union, intersect, minus operations. Aggregate Functions: avg, max, min, count, sum, Nested subqueries. Advanced SQL: Built-in Data Types in SQL, User-Defined Types, Large-Object Types.	SQL set operations and aggregates explained with examples; nested queries demonstrated step-by-step using multimedia; advanced SQL types introduced with short examples.	Final Exam	CO2
7	Integrity Constraints: Constraints on a Single Relation, Not Null constraint, Unique constraint, The check clause.	Integrity constraints taught through conceptual examples; real case scenarios discussed interactively; short practice questions reviewed in class.	Final Exam	CO2
8	Entity-Relationship Model: introduction, entity, entity set, attributes, E-R model constraints. Design issues of E-R model, Strong and weak entity sets, Cardinality, Relationship and relationship set, degree of relationship.	ER model introduced with diagrams; strong/weak entities explained with examples; students discuss sample ER structures.	Final Exam, Class Test 2	CO1

9	Extended E-R features: specialization, generalization, aggregation, design E-R diagram. Normalization: introduction, 1NF, 2NF, 3NF, 4NF, BCNF, Trivial, function and multi-valued dependencies.	Extended ER features taught using diagrams; normalization (1NF–BCNF) explained through table examples; functional & multivalued dependencies introduced with examples.	Assignment, Final Exam	CO1
10	Application Design and Development: triggers, Function, Cursor, Exception Handling and authorization in SQL. Application Security: Encryption techniques, Authentication, digital signature, digital certificates. SQL Injection, Encryption	Triggers and authorization explained with multimedia; security topics taught using real-world examples; students share examples of authentication/encryption.	Final Exam	CO2, CO3
11	Storage Structure: Overview of Physical Storage Media, Classification by capacity, speed and cost, Classification by type of storage, Classification by storage volatility, Stored Management. Magnetic Disks: Physical Characteristics of a Magnetic Disk, Performance Measures of Disks. RAID: Redundant Arrays of Independent Disks, RAID levels.	Physical storage media explained using diagrams; magnetic disk structure discussed using multimedia; RAID levels demonstrated through visual comparison	Final Exam	CO2, CO3
12	File Organization: Fixed-Length Records, Variable-Length Records, Organization of Records in Files, Indexing Basic Concepts, Ordered Indices, dense and sparse indices, B+ tree index files.	File organization types explained with examples; indexing concepts discussed using multimedia; B+ tree structure demonstrated with diagrams.	Final Exam	CO2, CO3
13	Hashing: types of hashing, hash file organization, static vs dynamic hashing, open vs closed hashing, indexing vs hashing. Query Processing: Overview, query-processing steps, Measures of Query Cost.	Hashing techniques explained using examples; indexing vs hashing compared through discussion; query processing and cost measures explained with examples.	Final Exam	CO2, CO3
14	Transaction State Diagram, Deadlock and Recovery Overview of Distributed and Cloud Database NoSQL Database	Transaction state diagram explained step-by-step; deadlock scenarios discussed through multimedia; prevention and avoidance methods reviewed interactively.	Final Exam	CO1, CO2, CO3

Part D: Assessment Approach

19. Assessment and Evaluation

Assessment Components		Marks Distribution
Continuous Internal Assessment (CIA) (50%)	Class Attendance/Class Performance	10%
	Class Test/Assignment/Quiz/Surprise Test	20%
	Mid Term	20%
Final Exam (FE)		50%
Total		100%

20. Assessment Pattern

1. Continuous Internal Evaluation (50 Marks)

*The percentage distribution of Bloom's categories in the assessment tools may vary by $\pm 5\%$.

Bloom's Category	Class Test 1/ Assignments 1	Mid Term Exam	Class Test 2/ Assignments	Attendance & Class Participation
Marks	(10)	(20)	(10)	(10)
Remember		2%		
Understand	5%	6%		
Apply	5%	8%	5%	
Analyze		4%		
Evaluate				
Create			5%	

2. Final Exam Evaluation (50 Marks)

*The percentage distribution of Bloom's categories in the assessment tools may vary by $\pm 5\%$.

Bloom's Category	Final Term (50)
Remember	

Understand	10%
Apply	20%
Analyze	10%
Evaluate	5%
Create	5%

21. Grading system

Numerical Grade	Letter Grade	Grade Point
80% and above	A + (A plus)	4.00
75% to less than 80%	A (A regular)	3.75
70% to less than 75%	A – (A minus)	3.50
65% to less than 70%	B + (B plus)	3.25
60% to less than 65%	B (B regular)	3.00
55% to less than 60%	B – (B minus)	2.75
50% to less than 55%	C + (C plus)	2.50
45% to less than 50%	C (C regular)	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Part E: References

22. Learning Materials

- 1) **Recommended Readings:** Lecture notes and Necessary documents (Uploaded in Google classroom).

- 2) **Supplementary Readings**

Text Book:

1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2006). Database system concepts (5th ed.). McGraw-Hill.
2. Rob, P., & Coronel, C. (2003). Database systems: Design, implementation, & management (5th ed.). Thomson Course Technology.
3. Elmasri, R., & Navathe, S. B. (2005). Fundamentals of database systems (5th ed.). Addison-Wesley.

Others (Site/Blog):

1. DBMS Tutorial – Learn Database Management System, Retrieved from: <https://www.geeksforgeeks.org/dbms/> (Accessed 22 Apr, 2025)
2. DBMS Tutorial, Retrieved from: <https://www.tutorialspoint.com/dbms/index.htm>
- 3) **Others.....**(as applicable for the discipline/academic program): N/A

Part F: Additional Information

1. Program Outcome: <https://uits.ac.bd/public/cse>
2. Outcome Based Curriculum: https://uits.ac.bd/public/obe_curriculum_cse
3. Course Catalog: https://uits.ac.bd/public/course_catalog_cse
4. Faculty Information: <https://uits.ac.bd/public/faculty-members-cse>
5. Assessments Rubrics:
6. Details of Bloom Taxonomy, Knowledge Profile, Complex Engineering Problems and Complex Engineering Activities:
7. Make-up Exam Procedure:

8. Attendance Policy:
9. Academic Calendar: <https://uits.ac.bd/academic-calendar/>
10. Academic Policies:
11. Proctorial Rules (Code of Conduct): <https://uits.ac.bd/wp-content/uploads/2024/01/Code-of-Conduct.pdf>